

The Core Philosophy

Topic 1: Shifting from Prediction to Causality in Data Science

MODULE: A/B TESTING FOUNDATIONS

Beyond Prediction

Why Classical ML isn't enough for business strategy.

The Spurious Correlation Trap

Ice Cream Sales



$r = 0.92$

Sunburn Incidents



Strong Link

Conclusion: Does eating ice cream cause sunburns?

No. Both are driven by a third variable: **The Summer Heat.**

Predictive ML vs. Causal AB Tests

Predictive ML

Goal: Minimize prediction error on historical data patterns.

-  Passive Observation
-  Correlation-based
-  Descriptive / Predictive

Causal AB Testing

Goal: Measure the incremental impact of a specific change.

-  Active Intervention
-  Causation-based
-  Prescriptive / Decisive

What Defines Causality?

For **X** to cause **Y**, three conditions must be met:

- 1 **Temporal Precedence:** X must occur before Y.
- 2 **Covariation:** When X changes, Y also changes.
- 3 **No Spuriousness:** No third factor explains both.

Traps in Observational Data



Selection Bias

Users who choose to use a feature are inherently different from those who don't.



Reverse Causality

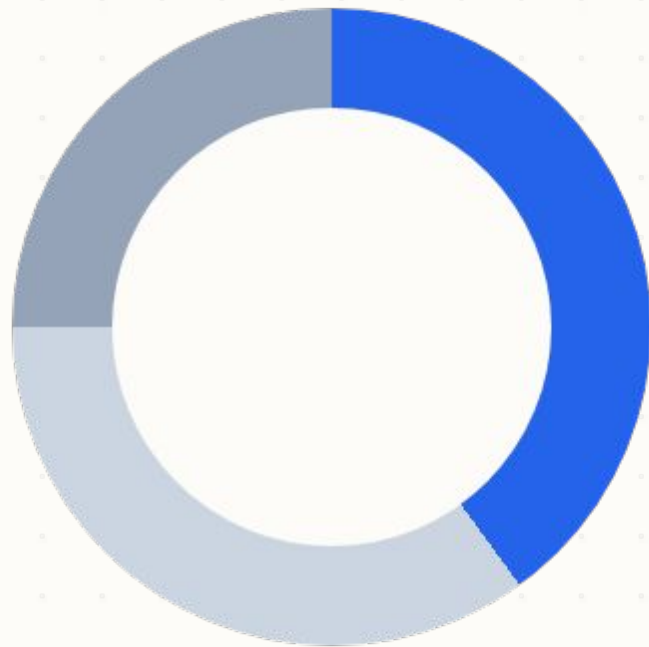
Did the app make the user active, or did active users find the app useful?



Confounding

External factors (holidays, competitors) driving both your change and your metric.

The Challenge of Data Bias



- **Direct Signal:** Real causal effect
- **Confounders:** Hidden external factors
- **Random Noise:** Statistical variance

The Counterfactual Framework

Counterfactual

What **would** have happened if they hadn't seen it.

$Y_i(0)$

Observed Outcome

What happened to the user after they saw the new design.

$Y_i(1)$

Fundamental Problem: We can never observe both for the same individual.

Why Accuracy Fails the Business

99.8%

Model Accuracy

The Precision Trap

A model can perfectly predict that "Users who use the search bar more are likely to purchase."

However, if you force every user to use the search bar, **purchases might decrease** if the search bar is annoying.

"Observation does not equal manipulation."

The Randomized Controlled Trial (RCT)

Randomization is our most powerful tool to "solve" causality by:

- ✓ **Breaking the Confounders:** Distributing all hidden traits equally between groups.
- ✓ **Ensuring Balance:** Creating "statistical clones" of your user base.
- ✓ **Isolating Impact:** The only systematic difference is your intervention.

Topic 2: Statistical Foundations

From Machine Learning to Hypothesis Testing

Managing risk and distinguishing signal from noise.

1. The Illusion of Difference

Why raw numbers can be misleading.

Did Variant B Actually Win?



The Problem of Sampling Variance:

If we flip a coin 100 times, we don't always get exactly 50 heads. That **0.3% lift** might be a genuine improvement, or simply **random noise**.

2. The Courtroom Analogy

Innocent until proven guilty.

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The Null (H_0)

Innocent until proven guilty.

The feature does **nothing**. Any difference is pure luck.

We assume this until evidence suggests otherwise.



The Alternative (H_a)

Proven Guilty.

The feature **caused** a change. Evidence suggests luck is a highly unlikely explanation.

3. Anatomy of a Decision

Mapping risk to a familiar ML concept.

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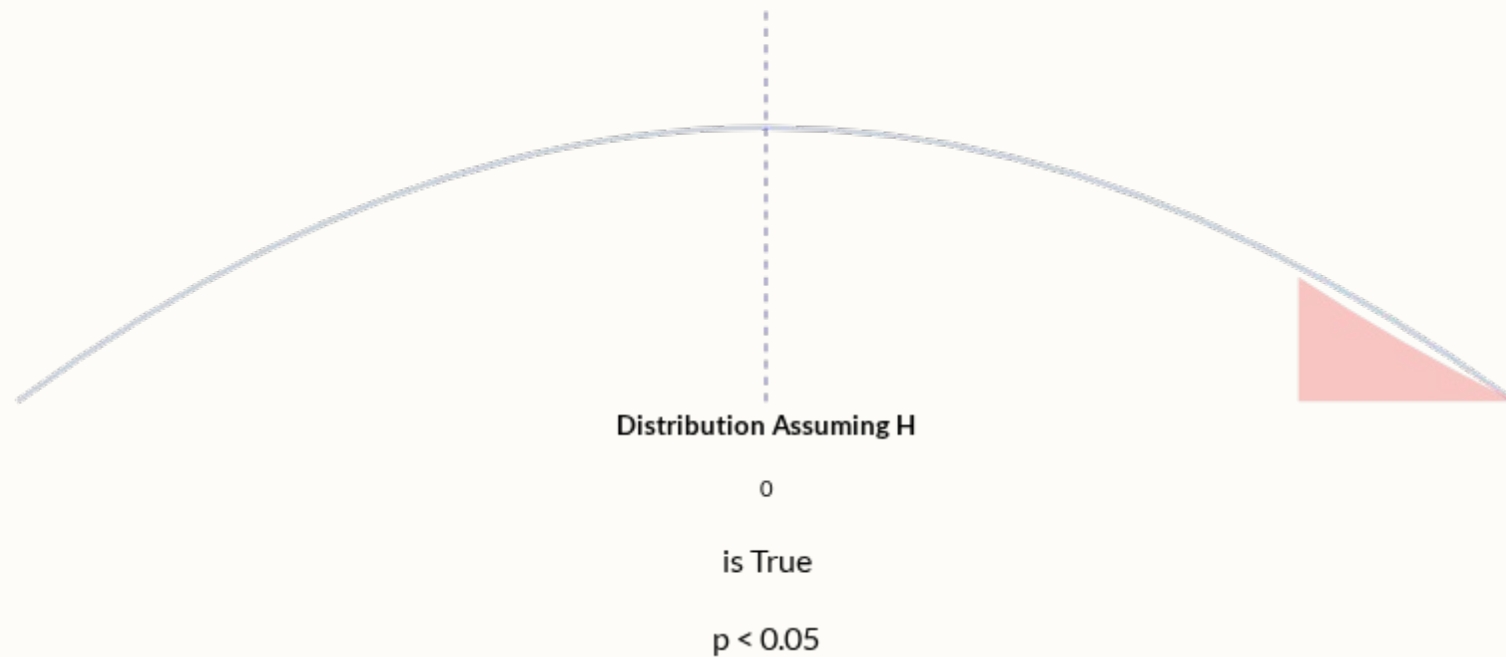
Reality	Decide: No Effect (H0)	Decide: Winner! (Ha)
H0 is True (Dud)	 Correct	Type I Error (α) False Positive
Ha is True (Winner)	Type II Error (β) False Negative	Power ($1-\beta$) Correct Detection

Mapping: This is exactly like the Confusion Matrix used in ML classification.

4. What is a p-value, Really?

The probability of "Luck."

4. What is a p-value, Really?



The probability of seeing a result **at least as extreme** as ours, assuming the feature does nothing. Low p-value =
High surprise = Reject the Null.

The Coin Flip Metaphor

Case A: 3 Heads

p-value approx 0.125 (12.5%)

Unusual, but happens often enough by chance. We **Fail to Reject** H_0 .



Case B: 10 Heads

p-value approx 0.001 (0.1%)

Extremely unlikely by luck. The coin is likely rigged. We **Reject** H_0 .



5. Significance vs. Effect Size

Bridging math to business value.

5. Significance vs. Effect Size

0.05%

Significant?

The Practical Check (MDE)

With enough data, tiny differences become "statistically significant" ($p < 0.05$), but they may be **practically useless**.

Minimum Detectable Effect (MDE): The smallest change worth tracking commercially. We only ship if the lift justifies the cost.

Topic 3: Metrics & Experiment Design

Defining success and planning the test duration before the first click.

A/B TESTING STRATEGY

Metric Frameworks

Choosing the right signals for decision making.

The Metric Hierarchy

Primary Metric

The North Star: The single metric that determines the winner of the experiment.

- ★ Direct link to business goal
- 🎯 Highly sensitive to change
- ✓ Examples: CTR, Conversion Rate

Secondary Metrics

The Context: Metrics that help explain *why* the primary metric changed.

- 🔍 Diagnostic information
- 🏗️ Funnel health checks
- ☰ Examples: Add-to-cart, Time on Page

Guardrail Metrics: "Do No Harm"



Technical Performance

Did the new feature increase page load latency or crash rates?



Business Health

Did we increase clicks but cause a massive spike in unsubscriptions?



Trust Metrics

Are we triggering more spam reports or customer support tickets?

Unit of Randomization

Ensuring consistent and unbiased user experiences.

How to Split Traffic?

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User ID / Cookie ID

Pros: Consistent experience across visits.

Cons: "Cookie churn" or users logging in on different devices.

Best for long-term behavioral changes.

Session ID

Pros: High volume of units; great for quick UI tests.

Cons: A user might see Version A now and Version B later tonight.

Risks "interference" and user confusion.

Sample Size Calculation

Planning the experiment duration (Power Analysis).

The 4 Pillars of Sample Size

Baseline Conv.

What is your current performance? (e.g. 5%)

Lower baseline = more users needed.

Significance (α)

Your risk tolerance for False Positives. (Default: 5%)

Lower α = more users needed.

Statistical Power

Probability of detecting a real effect. (Default: 80%)

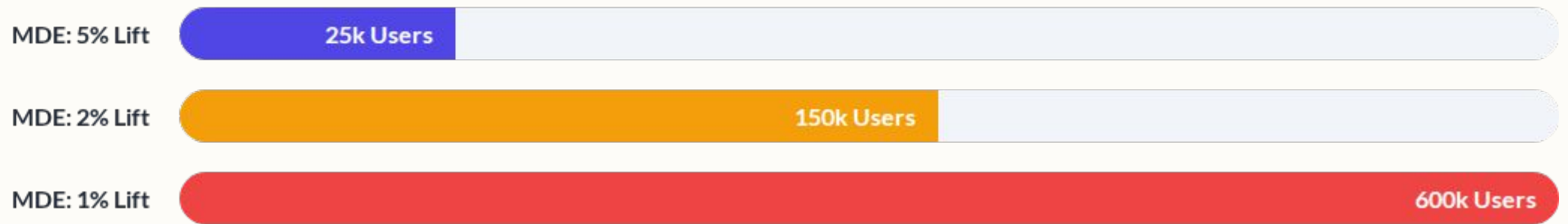
Higher power = more users needed.

MDE

Minimum Detectable Effect. Smallest lift you care about.

Smaller MDE = significantly more users.

The "Price" of Sensitivity



Trade-off: Searching for tiny improvements requires exponentially longer tests.

Practical Planning

14

Days (Min. Duration)

Why 14 Days?

Even if your sample size is reached in 2 days, you must capture a full weekly cycle (Day-of-week effect).

Tests usually run in multiples of 7 days to ensure seasonality doesn't bias the results.

The Math Behind the Plan

The required sample size per variant (n) is determined by:

$$n = \frac{16\sigma^2}{\Delta^2}$$

Δ = Minimum Detectable Effect
(MDE)

σ^2 = Variance of the metric

Notice: The sample size is inversely proportional to the square of the MDE!